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Physics-based forecasting of induced seismicity at Groningen gas field, the Netherlands: post hoc evaluation and forecast update

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Abstract

Induced seismicity due to natural gas extraction from the Groningen reservoir, the Netherlands, has been occurring since the early 1990s. Prospective forecasts of the possible maximum earthquake magnitude for different gas extraction scenarios have previously been published. However, their accuracy and continued relevance is unknown. Here, we evaluate our prior Groningen seismicity forecast and propose a revised model.

We compare the forecast to earthquakes that occurred after its publication, between 1 February 2017 and 1 July 2022. We used volume-scaling to account for actual gas extraction being 35% less than considered in the forecast. After this correction, we found that the forecast slightly underestimated the number of $M \geq 2.5$ events. However, it was consistent with the largest event in that period, an M 3.4. Then, we updated the forecast model for a new depressurization scenario, testing its pseudo-prospective (out-of-sample) performance using L- and N-tests. This revision considered two possible magnitude frequency distributions (MFD): (1) tapered Gutenberg-Richter (GR), where induced ruptures are confined within the reservoir, and (2) unbounded GR, where ruptures can propagate into the underburden. A tapered GR is the more likely scenario due to clear rollover in the MFD and a lack of events beneath the reservoir. However, we cannot exclude the possibility of an unbounded GR, because direct stress measurements in the underburden are not currently available. For the period 2021 to 2030, this new forecast suggests a 1% likelihood that the largest event will exceed M 3.8 for the tapered model, and M 5.4 for the unbounded model.

Keywords

Induced seismicity, Groningen, maximum magnitude, prospective forecast, post hoc evaluation.

Introduction

Around the world, hydrocarbon production and associated activities have led to sharp increases in seismicity, e.g., Oklahoma, USA [Keranen and Weingarten, 2018]; Alberta, Canada [Schultz et al., 2020]; and Groningen, the Netherlands [van Thienen-Visser and Breunese, 2015]. The seismic hazard arising from hy-

drocarbon production highlights the need to anticipate and manage induced earthquakes. But hazard assessments of induced seismicity face some unique challenges [Bommer et al., 2015, Bommer, 2022] as compared to standard hazard assessments for natural seismicity. For example, a recent re-evaluation of ground shaking in Oklahoma in 2018 [Petersen et al., 2018] revealed a higher chance of minor and moderate damage compared to some regions of California near the San Andreas Fault.

Standard probabilistic assessments of seismic hazard assume that the stochastic occurrence of earthquakes is governed by a stationary distribution [Baker et al., 2021]. The mean of this distribution is the regional earthquake rate, ideally constrained by a long earthquake catalog that averages out the stochasticity. However, for induced seismicity, the rate of earthquakes can vary dramatically over operational time scales: weeks-to-months for well stimulation [e.g., Häring et al., 2008, Schultz et al., 2020] and years-to-decades for reservoir production [Keranen et al., 2014, Majer et al., 2007, Muntendam-Bos et al., 2022]. To capture this year-on-year variability, the United States Geological Survey forecasts seismic hazard annually [Petersen et al., 2016, 2017, 2018] rather than attempting to compute a single, static hazard map. Apart from natural stochasticity, several production-related processes contribute to the short-term variability in triggered earthquake rate including pore-pressure diffusion [e.g., Hsieh and Bredehoeft, 1981, Shapiro et al., 2002, Dempsey and Riffault, 2019], poroelastic stresses [e.g., Chang and Segall, 2016, Goebel et al., 2017, Zhai et al., 2019], and hydraulic fracturing [e.g., Shapiro and Dinske, 2009, Schultz et al., 2020].

The prominence of hydrocarbon production as a cause of induced seismicity is both a challenge and an opportunity. It is a challenge because production data can be cumbersome, if not impossible, to assemble and model at a regional scale. But it is an opportunity because carefully considered operational proto-

cols could reduce the risk of triggered earthquakes [e.g., Dempsey and Riffault, 2019] and mitigate impacts on nearby communities. Realizing this opportunity requires models designed for specific hydrocarbon fields that connect operational protocols to probabilistic forecasts of the associated earthquake rate [e.g., Broccardo et al., 2017, Clarke et al., 2019, Li et al., 2022]. One example, and the subject of this study, is the prospective forecast model by Dempsey and Suckale [2017] of Groningen gas field in the Netherlands.

Once Europe's largest producing gas field, concerns about induced earthquakes have led to year-on-year reductions in Groningen's output from peak production in 2013. In 2016, the field operator, Nederlandse Aardolie Maatschappij (NAM), convened a workshop [Coppersmith et al., 2016] to solicit estimates of the maximum earthquake magnitude, $M_{\max}$, that could occur in the field [Bommer and Van Elk, 2017]. These estimates, of which the Dempsey and Suckale [2017] forecast was one contribution, were assessed by a separate expert panel who then developed a logic tree for $M_{\max}$. However, static models cannot capture inevitable deviations from planned production levels and evolving sociopolitical or economic drivers. Nor can such static models take advantage of new data or conceptual understanding to improve production plans. In 2022, NAM organized a second workshop to solicit updated estimates of $M_{\max}$ at Groningen and to develop a new logic tree [Coppersmith et al., 2022], following a similar structure to the 2016 workshop. The study described here was one contribution to the 2022 $M_{\max}$ workshop.

Our goal in this paper is to advance the state-of-the-art of induced-seismicity forecasting. We adopt an iterative development philosophy. We start by conducting a post hoc analysis of our prior forecasts in Dempsey and Suckale [2017] and then use this analysis as a basis to develop an improved model and a new prospective forecast of induced seismicity for Groningen gas. Our approach is

inspired by the value of adopting an inherently dynamic, so-called *living* model [Maldonado et al., 2022] to inform decision-making. Ideally, such models are subject to testing and accuracy checks [Schorlemmer et al., 2007, Király-Proag et al., 2016], post hoc evaluation [Strader et al., 2017, Dempsey et al., 2022] and improvement.

A prospective forecast is a probabilistic statement about a specific type of outcome (e.g., maximum earthquake magnitude, total number of earthquakes) happening within a well defined future time period. At Groningen, prospective forecasts have been published by Bourne et al. [2014], Dempsey and Suckale [2017], Bourne and Oates [2017a], Richter et al. [2020], Bourne and Oates [2020] and Limbeck et al. [2021]. A retrospective forecast, sometimes called a hindcast, fits a model to past seismicity data, hence building confidence that the model is accurately capturing seismogenic processes [e.g., Bourne and Oates, 2017b, Candela et al., 2019, Smith et al., 2022, Heimisson et al., 2022]. A third type of forecast is termed pseudo-prospective, wherein a forecast model is trained on one part of the available data (hindcasting) and then evaluated against the withheld (out-of-sample) portion [e.g., Bourne and Oates, 2017b, 2020, Richter et al., 2020, Limbeck et al., 2021, Smith et al., 2022]. This study makes use of all three forecast types.

Geological Setting

The 900 km^2 Groningen gas field is located in the northwest of the Netherlands, extending partially offshore (Fig.1a). The main gas-bearing layer is the Rotliegend Sandstone, located at an average depth of 3 km and varying in thickness from southeast to northwest [Bourne et al., 2014]. The reservoir is overlaid by Zechstein salt deposits up to 2 km thick and underlain by Carboninferous limestone. The reservoir is extensively faulted with groups of N-S, E-W, and

NE-SW trending normal faults (Fig.1a) with offsets ranging up to the thickness of the reservoir.

Natural gas has been produced from the Groningen reservoir since the 1960s. By July 2022, a total of 2250 billion cubic meters (bcm) had been produced, peaking at more than 80 bcm/yr in the late 1970s (Fig.1b). Production peaked again in 2013 at 54 bcm/yr but has since been declining. The mean reservoir pressure has declined more than 25 MPa from its initial value of 35 MPa (Fig.1c). Depressurization has been accompanied by mean reservoir compaction of 30 cm.

Induced seismicity at Groningen

Earthquakes were first detected in the Groningen field in 1991 (Fig.1b). A network extension in 1995 and further station deployment between 2014 and 2016 have steadily improved monitoring [Dost et al., 2017, Muntendam-Bos et al., 2022]. By July 2022, a total of 840 $M \geq$1.0 events have been detected in the field. The largest event to date was a M 3.6 in 2012 [Dost and Kraaijpoel, 2013]. While Spetzler and Dost [2017] and Willacy et al. [2019] have suggested that the majority of events were located within the reservoir, a later reanalysis by Smith et al. [2020] has suggested that a substantial proportion of events could have occurred in the overburden, specifically the Zechstein basal anhydrite. All three studies found little evidence for earthquake nucleation in the underburden at Groningen. However, underburden seismicity has been observed in other producing gas fields [Segall, 1989, Bardainne et al., 2008].

The nucleation depths of earthquakes is important, because it indicates what physical mechanism could be loading the faults. Earthquake nucleation within the reservoir can be explained by poroelastic stresses induced by depressurization [Segall and Fitzgerald, 1998, Dempsey and Suckale, 2017] or by compaction gradients that impose shear strains [Bourne and Oates, 2017b]. Earthquake nu-

cleation in the overburden can also be explained using poroelastic models with a different set of parameters [Smith et al., 2022]. Stress changes that favor nucleation may also locally promote rupture propagation [e.g., Dempsey et al., 2016, Buijze et al., 2017, Galis et al., 2017]. Ruptures whose dimensions are confined to the stress-perturbed region are commonly called "induced" to distinguish them from "triggered" ruptures that propagate without limit and may hence have a larger event magnitude.

One line of evidence for dimensionally-constrained earthquake ruptures is a relative deficit of large magnitude events in the magnitude frequency distribution (MFD) [Shapiro et al., 2011, Dempsey et al., 2016]. This effect is sometimes described as a rollover of the MFD at a corner magnitude – an event size where limited fault sizes or stress conditions within the field could restrict earthquake propagation. However, statistically confirming a rollover is difficult when only a few large events are in the catalogue [Main, 2000]. Rollover of the MFD can be modelled empirically using variants of the Gutenberg-Richter (GR) power-law, either by introducing a truncating maximum magnitude [Utsu, 1999] or an exponential taper [Bourne and Oates, 2020].

The MFD properties of Groningen earthquakes have been extensively studied. Bourne et al. [2014] analysed all $M \geq 1.5$ events between 1995 and 2012, and found that statistical uncertainty was too great to suggest evidence of a rollover or taper. They estimated a GR b-value between 0.90 and 1.19. Muntendam-Bos and Grobbe [2022] provided a more recent analysis and suggested there are variations in b-value across the field (between 0.8 and 1.5) but that these are stable in time. They also tested whether bounded GR distributions could describe the data and found evidence for a corner magnitude of 3.5 for a subset of events. In a subsequent study by Bourne and Oates [2020], a range of stress-dependent b-value and stress-dependent taper models were tested. Although

the stress-dependent taper had the best fit, the authors caution that a single model cannot be definitively preferred.

When the existence or nature of a taper is unclear, then a conservative alternative scenario is to consider an unbounded GR distribution. This approach implies that earthquake ruptures are not restricted to the reservoir and can propagate into underlying formations. We emphasize that this is not the most likely scenario for earthquakes at Groningen. For example, in addition to evidence for an MFD taper, Smith et al. [2020] estimated the likelihood that relocated hypocentres were below the reservoir to be less than 12%.

Prior models of induced seismicity

Several physical models have been proposed to explain earthquake occurrence at Groningen [Bourne et al., 2014, Dempsey and Suckale, 2017, Bourne and Oates, 2017b, Buijze et al., 2017, Candela et al., 2019, Smith et al., 2022, Heimisson et al., 2022]. Many assume that prior to gas production, ambient shear stresses within the reservoir were well below the critical limit of frictional failure [Jaeger et al., 2009, Zoback, 2010]. Evidence for this relatively low pre-stress is the 25 years and 1000 bcm of production at Groningen before the first earthquake was detected (Fig. 1b). An alternative explanation for the delay is that it reflects a long earthquake nucleation time scale that can be calibrated into a rate-and-state friction model. However, Heimisson et al. [2022] have found that such models struggle to capture the reduction in seismicity following its 2014 peak, and that a stress threshold is needed to correct the issue.

To explain induced seismicity at Groningen, models generally need to account for how gas production causes stress to accumulate on faults. The prevailing view is that depressurization of the reservoir has caused substantial compaction (Fig. 1c), which due to heterogeneity of material properties is spatially

variable across the field [Bourne et al., 2014]. Lateral gradients of compaction induce shear strains on faults embedded in the reservoir [Bourne and Oates, 2017b]. These strains are exacerbated on faults that offset reservoir components [Buijze et al., 2017, Candela et al., 2019].

An alternative model is that depressurization induces a poroelastic contraction within the reservoir, and this induces large extensional stresses [Segall and Fitzgerald, 1998, Dempsey and Suckale, 2017]. Smith et al. [2022] developed a more realistic poroelastic and compaction model that could estimate stress changes within and above the reservoir. They found that stress changes within the reservoir were stabilizing whereas changes above were destabilizing, which is consistent with their hypocenter inversion pointing to earthquakes in the overburden [Smith et al., 2020]. Other models have also considered whether rapid cycling of production wells could induce poroelastic transients that nucleate earthquakes [Segall and Lu, 2015] but were inconclusive and remain untested for Groningen [Postma and Jansen, 2018].

Once a suitable stress description has been selected, a model is needed for earthquake nucleation. A popular choice is the rate-and-state friction model of Dieterich [1994], which has few parameters. Candela et al. [2019] and Richter et al. [2020] have both applied this model to capture the onset and increase of seismicity at Groningen. A difficulty with this model is that it assumes faults begin as critically stressed, which is unlikely to be the case at Groningen. However, a modified rate-and-state formulation introduced by Heimisson et al. [2022] has addressed this issue.

Coulomb failure is an alternative description of earthquake nucleation. In this formulation, earthquakes nucleate when shear stress exceeds fault shear strength. Then, the seismicity rate, λ, is assumed to be proportional to the stress loading rate, i.e., $\lambda(t) \propto \dot{\tau} - f(\dot{\sigma}_n - \dot{p})$, where τ and σ_n are shear and

normal stresses, p is fluid pressure, and f is a friction coefficient. This model has been used by Bourne and Oates [2017b], Dempsey and Suckale [2017], and Smith et al. [2022] to provide an acceptable fit to the historical earthquake rate in Groningen.

Once an earthquake is nucleated, a model is needed to estimate its magnitude. A straightforward approach is to sample this directly from an assumed MFD for the field [e.g., Bourne and Oates, 2017b, Heimisson et al., 2022]. An alternative is to simulate the relevant fault friction and stress transfer physics [Dempsey and Suckale, 2017, Buijze et al., 2017], which can provide insights into limits on rupture dimension and hence MFD rollover. To be useful for forecasting, the simulation approach needs to be fast, which tends to favor either empirical models or highly reduced descriptions of earthquake propagation physics.

Methodology

Below, we summarize the particular model parameterizations we have used to approximate seismogenesis at Groningen. These include descriptions of the tectonic stress field, seismogenic faults, a poroelastic model that links gas production to fault loading, and simulations of source nucleation and size (Fig. 2). Additional model details can be found in Dempsey and Suckale [2017] or in the Electronic Supplement to this article. For brevity, we denote the Dempsey and Suckale [2017] study as DS17, and the updated model as DS22.

To characterize the pre-production stress, we assumed an extensional tectonic regime in which the vertical stress, σ_1, is given by depth integration of a constant overburden density (2500 kg/s) to the average reservoir depth (3 km). The minimum horizontal stress, σ_3, is rotated an angle θ clockwise from North. Its lowest possible value, σ_{crit}, is limited by brittle failure of the crust with fric-

tion coefficient, f [Jaeger et al., 2009]. Its actual value is parameterized by h, where $h = 0$ is a critically stressed crust and $h = 1$ is an isotropic stress state. The magnitude of the maximum horizontal stress, σ_2, is parameterized by H to be between the vertical and minimum horizontal stresses, with H=0 denoting $\sigma_2 = \sigma_3$ and H=1 denoting $\sigma_2 = \sigma_1$.

Summarizing, we impose a homogeneous pre-production stress fully described by four parameters: f, h, H, and θ. This controls the initial shear and normal stresses resolved on individual faults. We then impose further stress changes based on gas extraction and the associated pressure decline in the reservoir, $\Delta p(\mathbf{x}, t)$ (Fig. 2a). Depressurization has been modeled by NAM using a two-component (water and gas) reservoir simulator that accounts for historic extraction and has been history matched to pressure and subsidence data. Three future depressurization scenarios were provided by NAM as an input to the DS17 model, corresponding to extraction of 21, 27 and 33 bcm/yr between February 2017 and 2025.

We used the poroelastic model of Segall and Fitzgerald [1998] to link depressurization to reservoir stress changes. We approximated the reservoir as an Eshelby [1957] ellipsoidal inclusion that would contract as pressure drops were it not elastically coupled to the surrounding rock. Instead, the reservoir develops an extensional strain within and subsidence occurs above (Fig. 2c). We calculated the extensional strain locally via a poroelastic stress path coefficient, A [Segall and Fitzgerald, 1998, Zoback, 2010] and added this perturbation to the tectonic stress, $\Delta\sigma_h(\mathbf{x}, t) = A\Delta p(\mathbf{x}, t)$ where h=2 or 3.

Faults embedded in the regional stress field experience a time and space varying load (Fig. 2b). We have approximated faults as straight line elements. The ith fault has a normal vector, $\hat{\mathbf{n}}_i$, related to its strike, φ_i, and dip, ϕ_i, and a length, L_i. The evolving stress field causes normal, $\sigma_n(s, t)$, and shear stress,

$\tau(s,t)$, on the fault to vary with position and time (s is a coordinate along the strike of the fault). Our models included the 325 longest faults in Groningen (Fig. 1a), with a minimum fault length of 2 km.

In the DS17 model, the total poroelastic loading across all faults defined a time-varying seismicity rate for the field, $\lambda(t)$. Individual poroelastic loads were then imposed as inputs to a 1D fracture mechanics simulator that approximated earthquake nucleation and rupture on heterogeneous faults with constant vertical thickness. The output of this simulator was an ensemble of possible induced earthquakes for each fault, each with a different hypocenter, nucleation time, and magnitude larger than a minimum of M 1.0. Simulated earthquake were obtained for each of the 325 modeled faults and combined to create synthetic catalogs. Interactions between earthquakes on separate faults were not considered.

Improvements in the DS22 model

The major updates we have introduced are: (i) a revised depressurization scenario provided by NAM (see Electronic Supplement), (ii) representation of the faults' vertical dimension, which varies with reservoir thickness, (iii) replacement of the fracture-mechanics earthquake simulator with two empirical MFDs, (iv) model calibration that incorporates events locations, and (v) pseudo-prospective testing to estimate forecast accuracy.

Earthquake rate model

We define the rate density, ξ, as the seismicity rate per along-strike fault length. We assumed each fault had a time and space varying rate density of $M \geq 1.0$ seismicity, $\xi_i(s,t)$, which is proportional to the local stressing rate:

$$\xi_i(s,t) = B \times \dot{CFS}_i(s,t), \qquad \dot{CFS}_i(s,t) = \dot{\tau}_i(s,t) - f(\dot{\sigma}_{n,i}(s,t) - \dot{p}_i(s,t)), \quad (1)$$

where i denotes the fault index, s is an along-strike coordinate, CFS is the Coulomb Failure Stress that parameterizes the proximity of the fault to frictional failure, and $CFS > 0$ is the threshold condition for earthquake nucleation. The magnitude and evolution of stress components on a fault depends on the fault's orientation within the background stress field and its location within the depressurizing reservoir. The parameter B is a constant of proportionality determined later.

Faults in the DS22 model had a vertical dimension that was set by the thickness of the Groningen reservoir, increasing approximately linearly from southeast to northwest. This is important because faults in the thicker part of the reservoir have a larger surface area exposed to stressing. Hence, in our model, they have a higher rate of earthquake nucleation for the same stressing rate.

We incorporated reservoir thickness as a weight, $w(s)$, multiplied to each fault's local rate density, $\xi(s,t)$. We set the weight equal to one when the local thickness, $h(s)$, was equal to the reference thickness of 223 m, i.e.

$$w(s) = \frac{h(x(s), y(s))}{h(250, 592)}, \qquad h(x,y) = -3.04(x - 250) + 4.11(y - 592) + 223. \quad (2)$$

We derived this expression by fitting to the reservoir thickness map presented in Bourne and Oates [2017b].

A new depressurization scenario, $\Delta p(x,y,t)$, was provided by NAM for the construction of forecasts out to 2030. We applied this as a loading that produced

a rate density for the ith fault. We calculated a total seismicity rate for the fault by integrating its rate density over its length

$$\lambda_i(t) = \int_0^L \xi_i(s,t)ds. \tag{3}$$

The seismicity rate in a spatial bin can be calculated by modifying the integral limits in Eq. (3) to just that section of a fault within the bin.

We calculated the seismicity rate for the entire reservoir by summing the contributions of all faults

$$\lambda(t) = \sum_{i=1}^{325} \lambda_i(t). \tag{4}$$

We determined the constant of proportionality, B, by ensuring the total modeled seismicity over the training period, $[1960, t_{\text{train}}]$, matched the observed number of $M \geq 1.0$ earthquakes in that period, N_{train}

$$B = N_{\text{train}} / \int_{1960}^{t_{\text{train}}} \lambda(t)dt. \tag{5}$$

For pseudo-prospective tests, we set t_{train} to 1 February 2017 and N_{train} to 670. For the DS22 prospective forecast model, we set t_{train} to 1 January 2021 and N_{train} to 820.

We estimated the expected number of earthquakes, $\bar{N}_{\text{fcst}}$, in the forecast period, $[t_{\text{start}}, t_{\text{end}}]$, through a similar integral

$$\bar{N}_{\text{fcst}} = \int_{t_{\text{start}}}^{t_{\text{end}}} \lambda(t)dt. \tag{6}$$

For pseudo-prospective tests, we set t_{start} to 1 February 2017 and t_{end} to 1 July 2022, i.e., the DS17 evaluation period. For the DS22 prospective forecast, we set t_{start} to 1 January 2021 and t_{end} to 1 January 2030.

Synthetic catalogs and forecasts

A useful forecast product is the exceedance probability curve, which quantifies the likelihood that, in a defined time interval, there will be an earthquake whose characteristics exceed a certain level. We developed exceedance curves by considering the statistical properties of synthetic earthquake catalogs obtained from our model. Each catalog represents a possible future realization of seismicity at Groningen.

To construct a synthetic catalog, first, we calculated its size, N_{fcst}, by sampling a random integer from a Poisson distribution with mean, $\bar{N}_{\mathrm{fcst}}$. Thus, future catalogs have an uncertain size but with an average determined by the future seismicity rate, $\lambda(t)$. Second, for each event in the catalog, we assign it a magnitude, $M_i = [M_1, M_2, ...M_{N_{\mathrm{fcst}}}]$, by sampling from the earthquake simulator ensemble (for DS17) or an empirical MFD (for DS22). Third, we record the largest event in a catalog, $M_{\max} = \max(M_i)$, and the number of $M \geq$2.5 events, $N_{2.5} = \mathrm{count}(M_i \geq 2.5)$. Fourth, we construct 10^5 such catalogs and calculate a cumulative distribution function for the set of values $M_{\max}$ and $N_{2.5}$. These are the forecast exceedance curves.

Empirical magnitude model

The DS17 model could only represent earthquakes whose ruptures were confined to the reservoir layer. However, if earthquakes are able to rupture outside of the reservoir and into basement formations, then they could have larger magnitudes. In the DS17 model, this dimensional constraint manifested as rollover in the modelled MFD. Although this is consistent with observations at Groningen, excluding the possibility of underburden ruptures introduces uncertainty into our forecasts.

For DS22, we decided to generate earthquake magnitudes by randomly sam-

pling from an empirical MFD. We considered two possible MFDs: (1) tapered and (2) unbounded GR. They represent two approximate scenarios where earthquake source dimensions are either constrained to the reservoir or can rupture into the basement. We selected the distributions and their parameters from the Bourne and Oates [2020] study of Groningen seismicity. A detailed description can be found in the original article and the Electronic Supplement.

Model history-match and testing

Temporal history match

We history-matched our forecast models to the induced earthquake catalog maintained by KNMI (see Data and Resources). The DS17 model was calibrated to the event time information using a Markov-chain Monte Carlo (McMC) algorithm [Foreman-Mackey et al., 2013] to minimize model misfit. Model misfit was approximated by the log-likelihood (LLK) that scores a non-homogeneous Poisson process (NHP, the modeled seismicity rate, $\lambda(t)$) against a set of earthquake times [e.g., Bourne and Oates, 2017b, Dempsey and Riffault, 2019, Richter et al., 2020]

$$LLK_{\mathrm{nhp}} = \sum_{i=1}^{N} \log \lambda(t_i) - \int_{t_1}^{t_N} \lambda(t) dt. \tag{7}$$

Larger values of LLK_{nhp} denote progressively better fits of the model to the data. An advantage of LLK_{nhp} is that, when evaluated over parameter space, it can be interpreted as a parameter posterior. This in turn supports probabilistic sampling to create ensembles of seismicity rate models.

A drawback of temporal history matching is that it does not make use of the information in earthquake location data. There are advantages in requiring the model to also honour event locations. For instance, if certain fault orientations

dominate in parts of the field, then hypocenter data may provide clues about which fault orientations were inclined to fail first. This in turn can constrain parameters related to stress field orientation.

Spatio-temporal history match

For the DS22 model, we extended history-matching to account for both earthquake times and locations. To do this, we quantified the model misfit using a modified Gaussian log-likelihood. This approach has also been used by Smith et al. [2022] and Heimisson et al. [2022] to quantify misfit of Groningen seismicity models. The Gaussian LLK over N observations is written, up to a constant:

$$LLK_{\mathrm{gs}} = -\sum_{i}^{N} \frac{(\hat{x}_i - x_i)^2}{2\sigma_i^2} \tag{8}$$

where $\hat{x}_i$ is an observation, x_i is the model approximating the observation, and σ_i is the uncertainty associated with the observation.

Our model misfit, S, was comprised of four weighted terms associated with misfits to: (1) S_{t0}, the modelled event rate, λ_t, from the beginning of production in 1960 to just before peak seismicity in 2012, (ii) S_{t1}, modelled event rate during peak seismicity and the post-2014 decline, (3) S_x, modelled spatial event density, ρ_x, in the x direction, and (iv) S_y, the event density in the y direction. A weighting term, ω, allowed for relative prioritisation of the temporal and spatial matches.

$$
\begin{aligned}
S &= \omega(S_{t0} + S_{t1}) + (1-\omega)(S_x + S_y), \qquad (9)\\
S_{t0} &= -\frac{1}{51}\sum_{1960}^{2011}\frac{(\hat{\lambda}_t - \lambda_t)^2}{2\hat{\lambda}_t},\\
S_{t1} &= -\frac{1}{9}\sum_{2012}^{2021}\frac{(\hat{\lambda}_t - \lambda_t)^2}{2\hat{\lambda}_t},\\
S_x &= -\frac{1}{20}\sum_{1}^{20}\frac{(\hat{\rho}_x - \rho_x)^2}{2\hat{\rho}_x},\\
S_y &= -\frac{1}{20}\sum_{1}^{20}\frac{(\hat{\rho}_y - \rho_y)^2}{2\hat{\rho}_y}.
\end{aligned}
$$

We constructed the event rate observations, $\hat{\lambda}_t$, by binning all $M \geq 1.0$ seismicity at one year intervals. We constructed the event density observations, $\hat{\rho}_i$ where $i = [x, y]$, by binning all seismicity at either 2.1 km intervals in the x direction or 2.5 km intervals in the y direction. This corresponded to 20 bins in both directions across the study area. We assumed events in each bin were generated by a local Poisson process whose associated variance was equal to the observation, hence $\sigma_i^2 = \hat{\lambda}_t$ or $\hat{\rho}_i$. We set a minimum of variance of 1 to avoid divide by zero issues in bins with zero observations.

Although certain choices we have made in the construction of S may seem arbitrary, they are all practical decisions to improve model calibration. For instance, separating pre- and post-peak seismicity (S_{t0} and S_{t1}) made it easier for us to find models that capture this abrupt reversal in rate. In this study, we set ω to 0.75, which prioritised models that produced a good temporal match. However, within models with approximately similar temporal matches, we were still able to select those that also captured the spatial distribution. Our decision to separate out matches in time and two space dimensions was intentional to maximise the amount of data in each bin.

We calibrated the DS22 seismicity model using the Differential Evolution algorithm of Storn and Price [1997] as implemented in the Scipy Python library [Virtanen et al., 2020]. This is a genetic algorithm that minimizes a misfit function, in this case the expression S (Eq. 9). It does this by sampling parameters from a bounded space $[0 < h < 1, 0 < H < 1, 0° < \theta < 90°, 0.5 < f < 0.8, 0.8 < A < 1.0]$, saving the models with low misfit, and then recombining their parameters to search for improved fits. The optimization is global and was used with default parameter specifications and stopping criteria. The result is a large ensemble of possible model parameter combinations, each ranked by their misfit score.

Model testing

To measure the accuracy of our forecast model, we compare modelled and measured seismicity rates, either visually or quantified by a misfit function. In minimizing misfit, history-matching increases model accuracy over the training period, but does not guarantee the model's accuracy when comparing to future seismicity. To understand future performance, we test our model pseudo-prospectively by history-matching only a subset of the data and quantifying the model's accuracy against the withheld data. Schorlemmer et al. [2007] recommend two tests of forecast accuracy that can be used in prospective and pseudo-prospective settings. The L-test, which matches the modelled seismicity rate to the data, and the N-test, which checks for over- or underestimates of the number of earthquakes. Both tests were trialled by Candela et al. [2019] for hindcasting of a Groningen model.

The L-test considers whether the catalog of N event times, $[t_1, ...t_N]$, is consistent with a particular rate model, $\lambda(t)$. We score the model against the data using the NHP log-likelihood (Eq. 7). To evaluate model performance, Schorlemmer et al. [2007] proposed creating synthetic earthquake catalogs from

$\lambda(t)$, each with their own event times, $[\hat{t}_1, ...\hat{t}_M]$, where M could be different between catalogs. The synthetic catalogs were also scored with a likelihood. The distribution of synthetic scores provides a reference for evaluating the observed score, quantified as a quantile within this distribution. If the quantile is at the low end (e.g., <0.05), then the model provides a poor fit to the data; quantiles at the high end are acceptable [Schorlemmer et al., 2007]. In this study, the quantile is equivalent to 1-p where p is a probability along the exceedance curve.

In contrast, the N-test checks whether the total number of earthquakes forecast by a model for a given period is consistent with the number of earthquakes that actually occurred, N. This is achieved by comparing N to the distribution of catalog sizes, M, for the many synthetic catalogs constructed for the L-test. A quantile of N at the low and high ends of this distribution corresponds to models that, respectively, are over- and underestimating the earthquake rate.

Results

Post hoc evaluation of DS17 forecast

Dempsey and Suckale [2017] describe three forecasts produced by the DS17 model by extrapolating over the period 1 February 2017 to 1 January 2025 (Fig. 3a). Forecast seismicity was driven by different rates of depressurization corresponding to gas extraction averaging 21, 27 and 33 bcm/yr. Synthetic catalogs were sampled and used to construct exceedance curves for (1) the number of $M \geq 2.5$ earthquakes (Fig. 3b), and (2) the largest magnitude event in the forecast period, $M_{\max}$ (Fig. 3c). For example, for the 21 bcm/yr scenario, there was a 50% likelihood of 14 or more $M \geq 2.5$ events over the forecast period with an $M_{\max}$ of 3.4 or higher. There was a 5% likelihood of 21 or more $M \geq 2.5$ events with an $M_{\max}$ of M 3.9 or higher.

There were several challenges in retrospectively evaluating the DS17 forecast. First, actual gas extraction between 1 February 2017 and 1 July 2022 averaged 13.7 bcm/yr, which is about 35% less than the lowest scenario originally considered (21 bcm/yr). Second, we conducted our evaluation for the first 5.5 years of the nearly eight-year forecast period. Finally, the DS17 forecast is a probability distribution over future outcomes specifying the number of $M \geq 2.5$ events or maximum event size. We only have one opportunity to observe the future, but one sample is not sufficient to confirm a probability distribution. At best, if the actual outcome sits well outside the forecast, we might question its quality.

We partly address the reduced extraction and forecast window issues through a scaling argument. It is clear that gas extraction at a larger volumetric rate results in a higher rate of earthquakes (Fig. 3a). An inspection of DS17 modelled earthquake rates in the period 2017 to 2020 shows that the three extraction scenarios at 21, 27, and 33 bcm/yr correspond to ensemble average earthquake rates of about 35, 45 and 55 $M \geq 1.0$ events per year. Thus, the relative earthquake rates are in proportion to the relative gas extraction rates. This assumption is the basis of the Zöller and Holschneider [2016] forecast.

If, in 2017, we had had the foresight to construct a forecast for 13.4 bcm/yr gas extraction over nearly 5.5 years (total extraction of 72.5 bcm), then evaluation would be straightforward. In practice, the closest we have is an eight year forecast at 21 bcm/yr (total extraction of 168 bcm). Applying a volume scaling argument, we should therefore expect the number of earthquakes to be about 43% of the 21 bcm/yr forecast given in DS17. This would correspond to a 50% likelihood of 6 or more $M \geq 2.5$ events and a 5% likelihood of 9 or more. The actual number of $M \geq 2.5$ events between 1 February 2017 and 1 July 2022 was 10 (Table 1), which places this outcome at about the 2% ex-

ceedance level of the volume-scaled DS17 forecast. This exercise is equivalent to the N-Test [Schorlemmer et al., 2007] and suggests the DS17 model has been slightly underestimating the rate of $M \geq 2.5$ events.

There are a few possible interpretations of this result. The first is that it can be attributed as a plausible, if rare outcome, within acceptable bounds of stochastic uncertainty. In other words, the earthquakes during this period were a 1 in 50 event and such outcomes do occur from time to time. In this case, the N-test is fair and the conclusion should be to accept the forecast. This is fine as a hypothesis, although it cannot be tested without constructing, waiting and retrospectively testing many more forecasts. That is not very practical, nor a particularly satisfying explanation.

The second explanation is that the volume scaling argument, although consistent with the narrow range of tested extraction scenarios (21 to 33 bcm/yr), is less accurate when extrapolated outside this range, to an effective rate of 13.4 bcm/yr. In this case, the N-test is not really fair because the forecast has been misrepresented by volume scaling. The third explanation is that a deficiency in the model had lead to an underestimate of $M \geq 2.5$ events in the forecast period. This could be due to limitations of the reservoir simulator that produced the depressurisation scenarios, the simple poroelastic model we used to convert pressure decline into stress changes, or in the earthquake source simulator itself. In this case, the N-test is fair and the conclusion should be to reject the forecast.

Another forecast output is the maximum event magnitude. Unlike $M \geq 2.5$ events, this quantity is not straightforward to rescale given known variation from the original scenarios, because $M_{\max}$ depends on the statistics of extreme value distributions. It is also influenced by any truncation at high magnitudes in the MFD, which is likely to be relevant for Groningen forecasts. Therefore,

the best argument we can offer here is a qualitative one.

Considering the $M \geq 2.5$ event forecast for the 21 and 33 bcm/yr extraction scenarios (Fig. 3b), the major parts of their respective exceedance curves span from 10 to 25 $M \geq 2.5$ events. Across this range, there is a difference of about 5 events between the two scenarios. In comparison, exceedance curves for the largest event are more similar across the scenarios, differing by about 0.05 magnitude units. We find that in the GR and truncated distribution, the largest events are also the rarest outcomes. An order of magnitude increase in the size of catalog is required to induce a magnitude increment in the largest event. Here, the observed catalog, comprising 10 $M \geq 2.5$ events, is of a similar size to the scenario medians in DS17 (between 15 and 20 events). For this reason, the largest event exceedance curves given in DS17 are likely to be only a marginally too-high representation of the 13.7 bcm/yr scenario. In this context, the maximum event size magnitude of M 3.4 in the evaluation period appears consistent with the DS17 forecast.

Updated DS22 forecast

The new forecast by our DS22 model accounts for revised depressurization, history matching to earthquake locations, a choice of two empirical MFDs, and pseudo-prospective testing.

Pseudo-prospective testing

We estimated the forecast skill of the DS22 model under pseudo-prospective conditions. First, we used the earthquake catalog prior to 1 February 2017 to train a minimum misfit seismicity model. This required that we reduce the upper time limit of the term S_{t1} in Eq. (9) from 2021 to 2017 and rebalance the weighting factor from 1/9 to 1/5. Then, we used data between 1 February 2017

and 1 July 2022 to evaluate the trained model.

Fig. 4(a) shows that the trained model replicates the onset of seismicity in the 1990s, growth through the early 2000s, peak in 2014 and then decline to 2017. Seismicity onset is late by about two years, but the pattern is a reasonable match thereafter. During the test period, seismicity declines steadily at a rate that appears consistent with the data. This model had a similar initial stress distribution to DS17 [see Dempsey and Suckale, 2017, fig 2(c-e)] but has a lower friction coefficient. Like the DS17 model, the poroelastic stress path parameter is higher than reported for real reservoirs. In our opinion, this deviation likely reflects missing physics in the model.

Fig. 5 presents modelled earthquake area density for the minimum misfit model we used for pseudo-prospective testing. The spatial distribution approximately captures two clusters of seismicity in the center and southwest of the field, although small underestimates of the event density are noted.

We have performed the L-test for withheld events in the test data set. The observed likelihood had a value of 460, which places it at the high end (4% probability to exceed) of the synthetic distribution of likelihoods (Fig. 4(b)). On this basis, the rate model can be judged consistent with the data.

The N-test is a second check on the rate model, in this case to check whether it is over- or underestimating the total number of events (Fig. 4(c)). We constructed exceedance curves for the number of $M \geq 2.5$ events for both MFD models in the evaluation period. The actual number of $M \geq 2.5$ events that occurred during this 5.5 year period was 10. This outcome had a 9 and 11% probability to exceed, respectively, for the tapered and unbounded GR models.

Finally, we considered the exceedance curve for the largest magnitude event (Fig. 4(d)). Both MFD models were consistent with the largest M 3.4 event that occurred during the evaluation period. This outcome had a 34 and 60%

probability to exceed, respectively, for the tapered and unbounded GR models.

Based on L-test, N-test and $M_{\max}$ exceedance test, we find no clear evidence to reject this model. Therefore, we have proceeded to use it for prospective forecasting.

Prospective forecast: 2021 to 2030

We constructed the prospective DS22 forecast by retraining to obtain a minimum misfit model using earthquakes up to 1 January 2021. The resulting model had very similar parameters to the one trained during pseudo-prospective testing (Table 2). The extrapolated seismicity rate in the period 2021 to 2030 shows a steadily declining seismicity rate, dropping from 20 to 10 $M \geq 1.0$ events per year.

For this model, we constructed synthetic catalogs for the forecast period and used these to define exceedance curves for both MFD scenarios. Both the tapered and unbounded GR models produced similar estimates for the number of future $M \geq 2.5$ events, with a 10% chance to exceed 8, a 1% chance to exceed 11, and a 0.1% chance to exceed 13 or 14 (Table 3).

In contrast, our estimate of $M_{\max}$ varies considerably between the two MFD models. The tapered GR model has a rollover at M 3.5, which exerts an upper constraint on $M_{\max}$ at very low exceedance probabilities, e.g., 0.1% chance to exceed 3.9. In comparison, the unbounded GR model has no such constraint, indicating larger $M_{\max}$ values of 4.3, 5.4 and 6.5 at 10, 1 and 0.1% exceedance.

An analysis of model sensitivity to key parameters has been included in the Electronic Supplement. We found that the model could achieve a reasonable match to the data even when one of its parameters was knowingly assigned an incorrect value. However, this has a minimal effect on forecast outputs, particularly $M_{\max}$, which are largely constrained by the history matching process.

Discussion

Possible maximum magnitude

Our goal in this study was to estimate the *possible* maximum magnitude of seismicity at Groningen. This is different from the *expected* maximum magnitude [Zöller and Holschneider, 2016]. We take the expected maximum magnitude as a distribution expectation, that is, the magnitude associated with the 50% exceedance probability on the cumulative distribution function for $M_{\max}$ (Fig. 6C or the top row of Table 3). For our tapered GR model, this is an M 3.2 event, and for our unbounded GR, it is M 3.5. These values are similar to those obtained by Bourne and Oates [2020] (their table 4), who tested a range of MFD models.

We take "possible" in possible maximum magnitude to mean an outcome with finite probability. In this sense, all values along the exceedance curve are possible maximum magnitudes, each occurring with increasing degrees of improbability. For our tapered GR model, possible magnitudes increase very little from 10 to 0.1% exceedance, from M 3.6 to 3.9. For our unbounded GR model, possible maximum magnitudes are higher, e.g., M 5.4 at 1% exceedance. This is at the top end of the range of the various taper models tested by Bourne and Oates [2020] for the same exceedance probability.

For our analysis, the choice between tapered or unbounded GR model is the most significant determinant of $M_{\max}$. We decided to include an unbounded MFD despite it being less likely for two reasons: First, it is possible that unbounded GR is the appropriate generating MFD for Groningen seismicity to date and the apparent absence of larger magnitudes is an unlikely statistical artefact. Previous studies [Bourne and Oates, 2020, Muntendam-Bos and Grobbe, 2022] have judged this situation improbable – we are inclined to agree – but none of them have confidently ruled out the unbounded GR. In the logic

tree constructed by the expert panel of the 2022 Mmax workshop [Coppersmith et al., 2022], a probability of 10% was assigned to the scenario that ruptures could occur within the underburden.

The second argument for considering an unbounded MFD is that, although tapered GR is likely a better description of seismicity to date, we cannot confidently assess that future seismicity will not later switch to an unbounded distribution. For example, deepening or widening of pressure depletion could impose a larger poroelastic loading across a greater underburden volume [Segall, 1992]. Ultimately, such critical uncertainties are best investigated with direct observational evidence such as drilling into the underburden, performing pressurization tests and borehole image analysis to determine in-situ stress conditions, or obtaining drill-core to test for velocity-weakening frictional properties. Until these data are obtained it is, in our view, unwise to categorically rule out the possibility of unbounded ruptures.

Other definitions of the possible maximum magnitude appeal to inherent physical limits of the seismogenic system. For example, Bourne et al. [2014] estimated an $M_{\max}$ of 6.5 corresponding to all induced compaction strain were released in a single event. The longest fault included in the DS17 model could potentially generate an $M_{\max}$ of 6.4, although this never happened in the very large number of catalogs generated in that study. Compared to $M_{\max}$ defined probabilistically, it is less clear how to estimate relative likelihoods for these competing models. An expert-led process that estimates weights in a logic tree [Bommer and Van Elk, 2017] is one way to do this, e.g., the deliberations of the 2022 Mmax expert panel that assigned a 10% weight to the unbounded-rupture branch [Coppersmith et al., 2022].

Correctness of model physics

The models we have presented do not include all possible physics relevant to induced seismicity at Groningen. Arguably, the major physical shortcoming has been our decision not to incorporate differential compaction in the reservoir as an additional load on faults [e.g., Bourne et al., 2014, Bourne and Oates, 2017b, Richter et al., 2020]. Incorporating this effect in future updates has the potential to improve the model's spatial match to seismicity because compaction has a different pattern than pressure decline [e.g., Bourne et al., 2014, fig. 9].

Our study, as well as those of Bourne and Oates [2017b] and Smith et al. [2022] have used Coulomb stressing rates as a proxy for earthquake nucleation rather than implementing a rate-state nucleation model as others have done [Candela et al., 2019, Richter et al., 2020, Heimisson et al., 2022]. Both our model and prior studies [Bourne and Oates, 2017b, Smith et al., 2022] produce a reasonable match to the data, which means model misfit alone is a poor basis for preferring one set of physics over the other. Rate-and-state models may struggle to replicate the sharp decline in seismicity after 2014 (e.g., Candela et al. [2019] fig. 6 and Richter et al. [2020] fig. 9), although this may have been solved with modified formulations that introduce stress thresholds [Heimisson et al., 2022].

Despite the different choices available for constructing seismicity rate models, it is not clear these will have a major impact on the output forecast for $M_{\max}$. This is because, in fitting to the same data, all such models are taught to capture first-order features of the seismicity time series: its onset, rise, peak, and decline. Thus, the historical context exercises quite a strong constraint on their future behaviour. Ultimately, we expect that minor differences in modelled earthquake rates arising from different physical formulations will be small compared to the uncertainty in the MFD.

We emphasize that our study is one of several published forecast models for induced seismicity at Groningen [Bourne et al., 2014, Zöller and Holschneider, 2016, Bourne and Oates, 2017b, Shapiro et al., 2017, Candela et al., 2019, Richter et al., 2020, Limbeck et al., 2021, Smith et al., 2022, Heimisson et al., 2022]; all combining different physics and forecasting methods. Collectively, these can be viewed as a model ensemble with certain conferred advantages [Marzocchi et al., 2012, Taroni et al., 2014]. For quantities such as the post 2014-peak seismicity rate, several of the forecasts have developed consensus, which outlier models can use as a reference to evaluate their own methods. At the same time, the consensus has an opportunity to observe the outliers and consider whether they are revealing any particular blindspots in the conceptualization of the majority.

Conclusion

We have undertaken a post hoc evaluation of the Dempsey and Suckale [2017] Groningen seismicity forecast (termed DS17) for the period 1 February 2017 to 1 July 2022. We have also developed a new prospective forecast for 2021 to 2030 (termed DS22). Our goal with both forecast models was to estimate an exceedance probability curve for the largest event in the forecast period, $M_{\max}$, using a physics-based model that links gas extraction and depressurization to earthquake nucleation on reservoir faults.

Our updated DS22 model incorporated earthquake locations and a new depressurization plan, and performed well in out-of-sample (pseudo-prospective) testing. A critical uncertainty that remains was whether future induced ruptures would be dimensionally bounded by the reservoir. Studies to date [Bourne and Oates, 2020, Smith et al., 2020, Muntendam-Bos and Grobbe, 2022, Coppersmith et al., 2022] have concluded that rupture into underlying Carboniferous

limestone is unlikely, but none have ruled it out as a possibility. This uncertainty is particularly consequential to the forecast, with a 1% probability to exceed M 3.8 for bounded ruptures and M 5.4 for unbounded, between 2021 and 2030. We adopt the consensus view that bounded ruptures ($M_{\max}$ 3.8) should be the preferred model. We recommend that direct observation and testing of the underburden could be a practical approach for reducing this uncertainty.

Data and Resources

The earthquake catalog used in this study can be obtained from KNMI at `http://cdn.knmi.nl/knmi/map/page/seismologie/all_induced.csv`. Gas extraction volumes can be obtained from NAM at `https://www.nam.nl/gas-en-olie/gaswinning.html`. The depressurization scenario used in the updated forecast has been included in the Electronic Supplement. All other data are available from cited references in the text. The code needed to recreate the results of this study is available at `https://doi.org/10.5281/zenodo.7109337`. The full dataset made available to participants of the Mmax workshop is available at `https://public.yoda.uu.nl/geo/UU01/RHHRPY.html`. Supplemental Material for this article includes additional details about the seismological model and a sensitivity analysis.

Declaration of Competing Interests

This research was completed and published under contract to NAM as a contribution to the Second Workshop on Mmax for Seismic Hazarad and Risk Analysis in the Groningen gas field.

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Table 1: Gas extraction and seismicity over the evaluation period.

Period	Extraction (bcm)	Seismicity N(M≥1.0)	$M \geq 2.5$ events
2017, 1-Feb to 31-Dec	21.2	53	M 2.6 (27/05/17)
2018, 1-Jan to 31-Dec	18.8	28	M 3.4 (08/01/18) M 2.8 (13/04/18)
2019, 1-Jan to 31-Dec	15.6	35	M 3.4 (22/05/19) M 2.5 (09/06/19)
2020, 1-Jan to 31-Dec	7.9	34	M 2.5 (02/05/20) M 2.7 (14/07/20)
2021, 1-Jan to 31-Dec	6.5	29	M 2.5 (04/10/21) M 3.2 (16/11/21)
2022, 1-Jan to 31-Jun	2.5	4	M 2.7 (01/04/22)

Table 2: Summary of forecast model parameters. DS17=Dempsey and Suckale [2017], DS22=this study, PRO=prospective, PPRO=pseudo-prospective.

Parameter	DS17 PRO	DS22 PPRO	DS22 PRO
h	[0.16,0.26]	0.19	0.19
H	[0.2,0.45]	0.36	0.38
θ	[25°, 34°]	46.0°	46.6°
f	[0.50,0.72]	0.50	0.50
A	[0.90,1.0]	1.0	1.0

Tables

Table 3: DS22 forecast and select exceedance probabilities for the period 2021 to 2030. $N_{2.5}$ values are rounded to the nearest integer.

p	$N_{2.5}$		$M_{\max}$	
	tapered	unbounded	tapered	unbounded
0.5	5	5	3.2	3.4
0.1	8	8	3.6	4.3
0.01	11	11	3.8	5.4
0.001	13	14	3.9	6.5

Figure 1: Overview of Groningen field and gas production activities. (a) Map of Groningen with field outline (black), straight line approximations of faults (red), and induced earthquakes. (b) Annual gas extraction (black line) in billions of cubic meters (bcm) and annual induced earthquakes (open circles) with $M \geq 1.0$. (c) Average reservoir pressure (solid line) and compaction (dotted) over time.

Figure captions

Figure 2: Model summary. (a) Fault of length, L, represented as a (red dashed) 1D line element with strike, φ, and along-strike coordinate, s. Contours of pressure decrease (MPa) at 2021 are shown in blue. Blue arrows in top corner show one possible orientation of the two horizontal principle stresses rotated an angle, θ, from north. (b) 3D view of fault showing dip angle, ϕ, and resolved normal and shear stress components, σ_n and τ, that vary along-strike and with time. In DS17, fault vertical thickness was a constant, whereas in DS22 thickness varies along strike. (c) Poroelastic inclusion model adapted from Segall and Fitzgerald [1998] showing how (i) for a non-elastically coupled reservoir, depressurization induces a contraction, (ii) elastic coupling is restored by stretching the contracted reservoir back to its initial volume, and (iii) subsidence compensates the vertical stress change leaving only an extensional horizontal stress in the reservoir. (d) Example rupture length simulation for calculation of event magnitude in the DS17 model, adapted from Dempsey et al. [2016]. Crack-like propagation of slip along the 1D fault, arresting in sections of negative stress drop.

Figure 3: Evaluation of DS17 forecast. (a) DS17 seismicity rate model ensemble showing best fits (black lines) through data (open circles) and three future scenarios: 21 (green), 27 (orange) and 33 bcm/yr (purple). Modelled and observed MFD is shown in the inset. Horizontal dashed lines show approximate scenario median rates between 2017 and 2020. (b) Exceedance probability for number of $M \geq 2.5$ events in the period 1 February 2017 to 1 January 2025 (solid lines) and volume-scaled scenario for 21 bcm/yr (dashed line) for the evaluation period 1 February 2017 to 1 July 2022. Actual number of $M \geq 2.5$ events is shown by the vertical dotted line. (c) Exceedance probability for the largest event in the forecast period for the three scenarios. Actual largest event in the evaluation period shown by the vertical dotted line.

Figure 4: Pseudo-prospective performance of the updated model over the evaluation period. (a) Measured seismicity rate (circles) compared to best fit model, $\lambda(t)$ (solid line) with 90% aleatoric variability (grey envelope). The data are divided into train and test sets at the year 2017 (closed and open circles, respectively), with the former used to calibrate the model shown in Fig. 5. The trained model is extrapolated for the period February 2017 to July 2022 and compared to the withheld test data (open circles). The greyed marker is the extrapolated event count for 2022 based on events in the first six months of the year. (b) Exceedance curve for likelihoods scoring $\lambda(t)$ against synthetic catalogs derived from $\lambda(t)$. Likelihood of $\lambda(t)$ scored against the data is shown by the vertical dotted line (LLK=460, p=0.04). The (c) Exceedance probability curve for the number of $M \geq$2.5 events, and (d) the largest magnitude event, for tapered (solid) and unbounded (dashed) Gutenberg-Richter distributions. Actual outcomes during the evaluation period are shown as vertical dotted lines ($N_{2.5}$=10 with p_{tp}=0.09 and p_{ub}=0.11; $M_{\max}$=3.4 with p_{tp}=0.34 and p_{ub}=0.60; tp=tapered and ub=unbounded).

Figure 5: Model fit to spatial seismicity distribution at Groningen. (a) Model match (solid line) to event density (circles) in 2.1 km bins along the x direction. The grey shaded envelope shows an interval of 90% aleatoric variability on the modelled density. (b) Heat map showing area density of cumulative model seismicity as at February 2017 in 2.1×2.5 km bins. Also shown are modelled faults (straight lines) and $M \geq$1.0 events (open circles). (c) Model match to event density in 2.5 km bins along the y direction.

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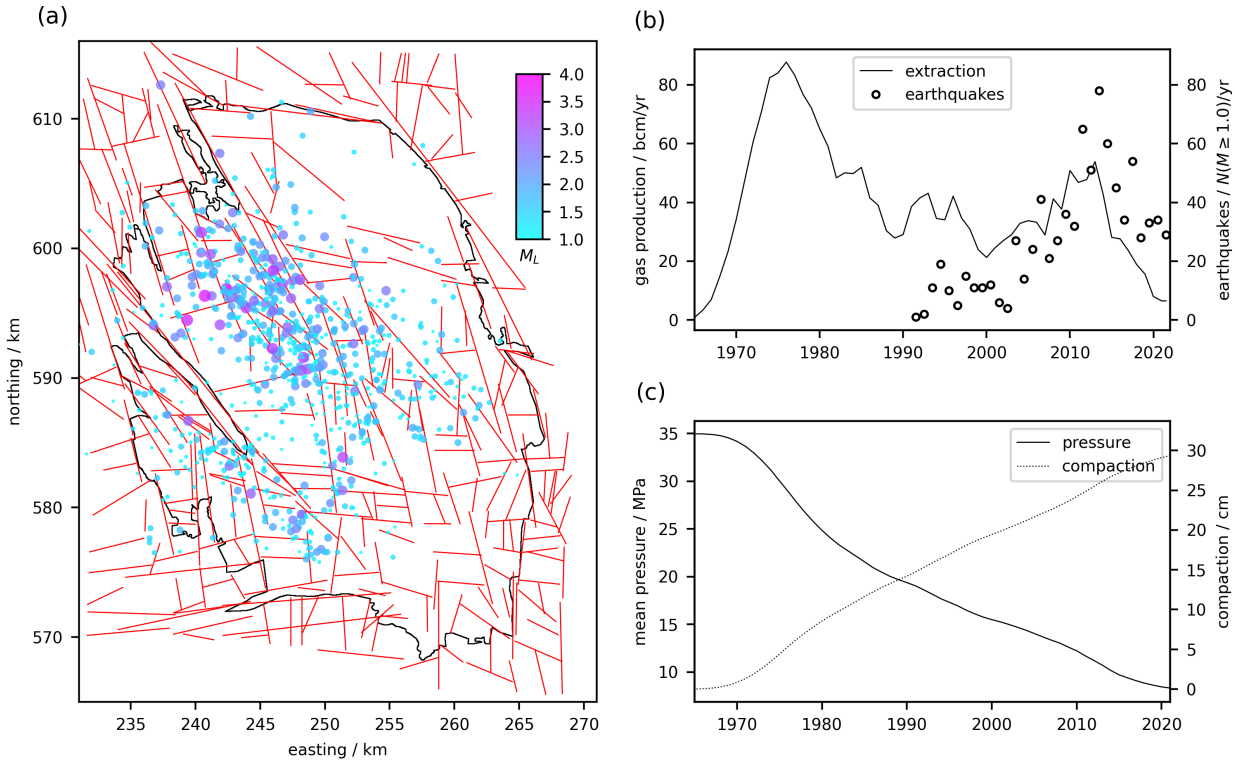


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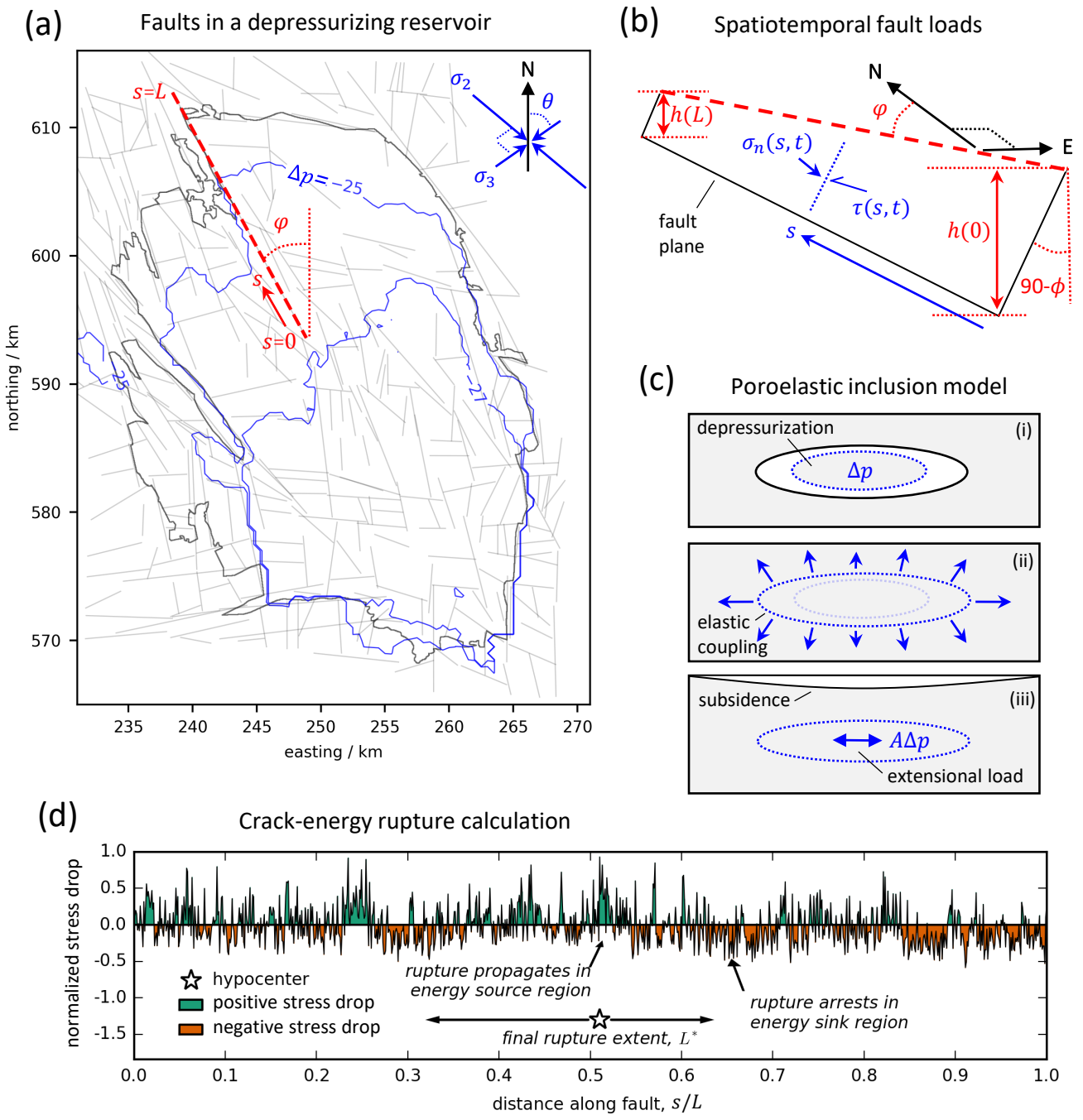


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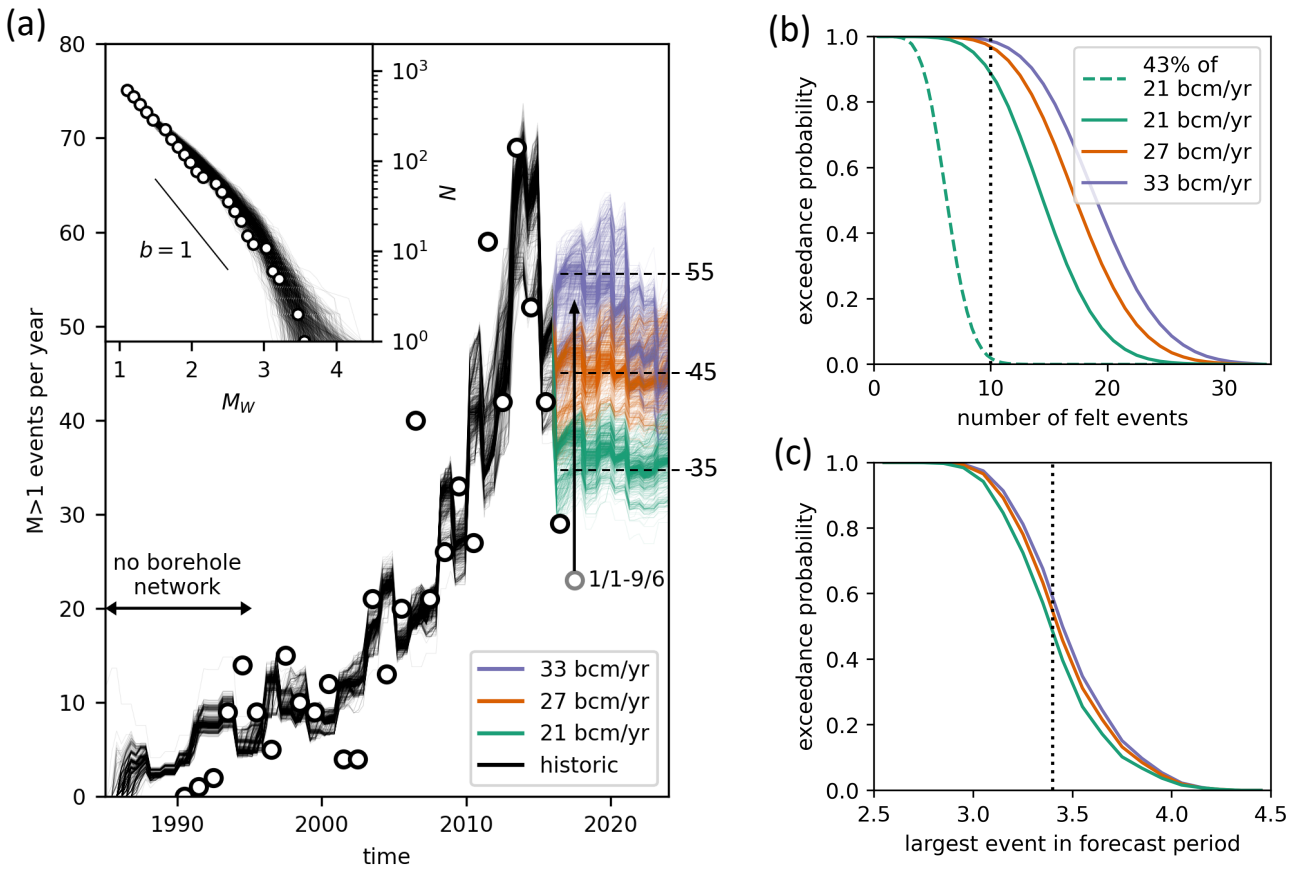


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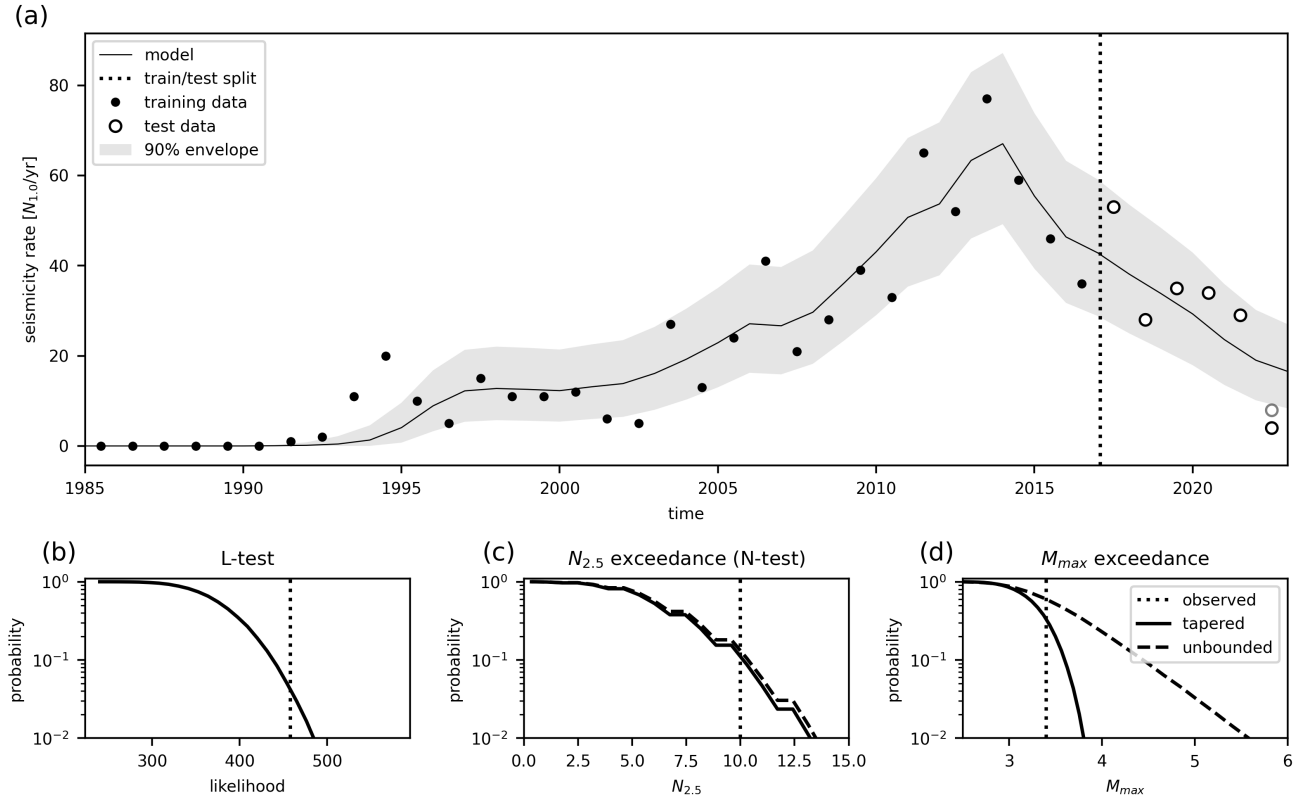


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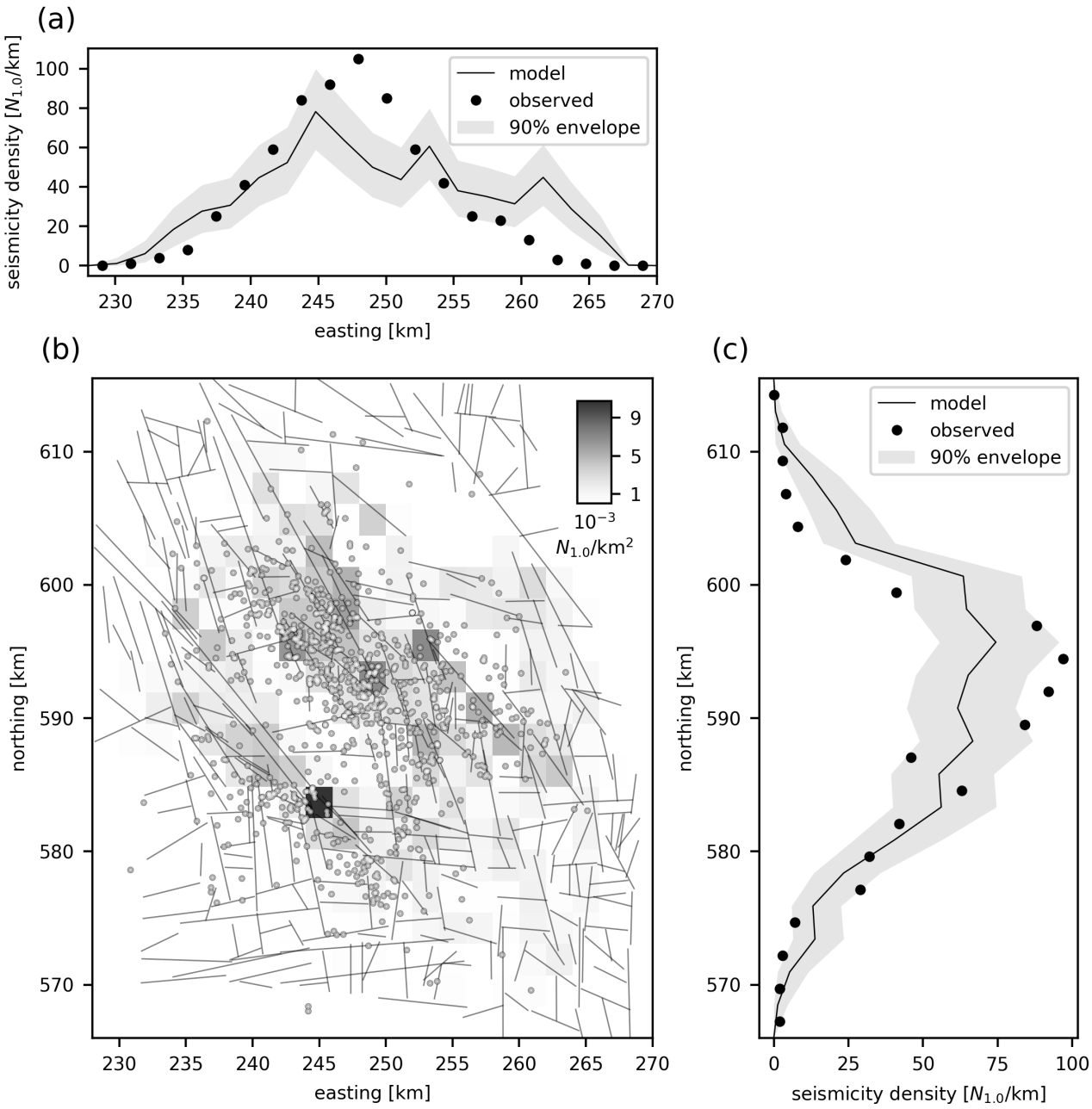


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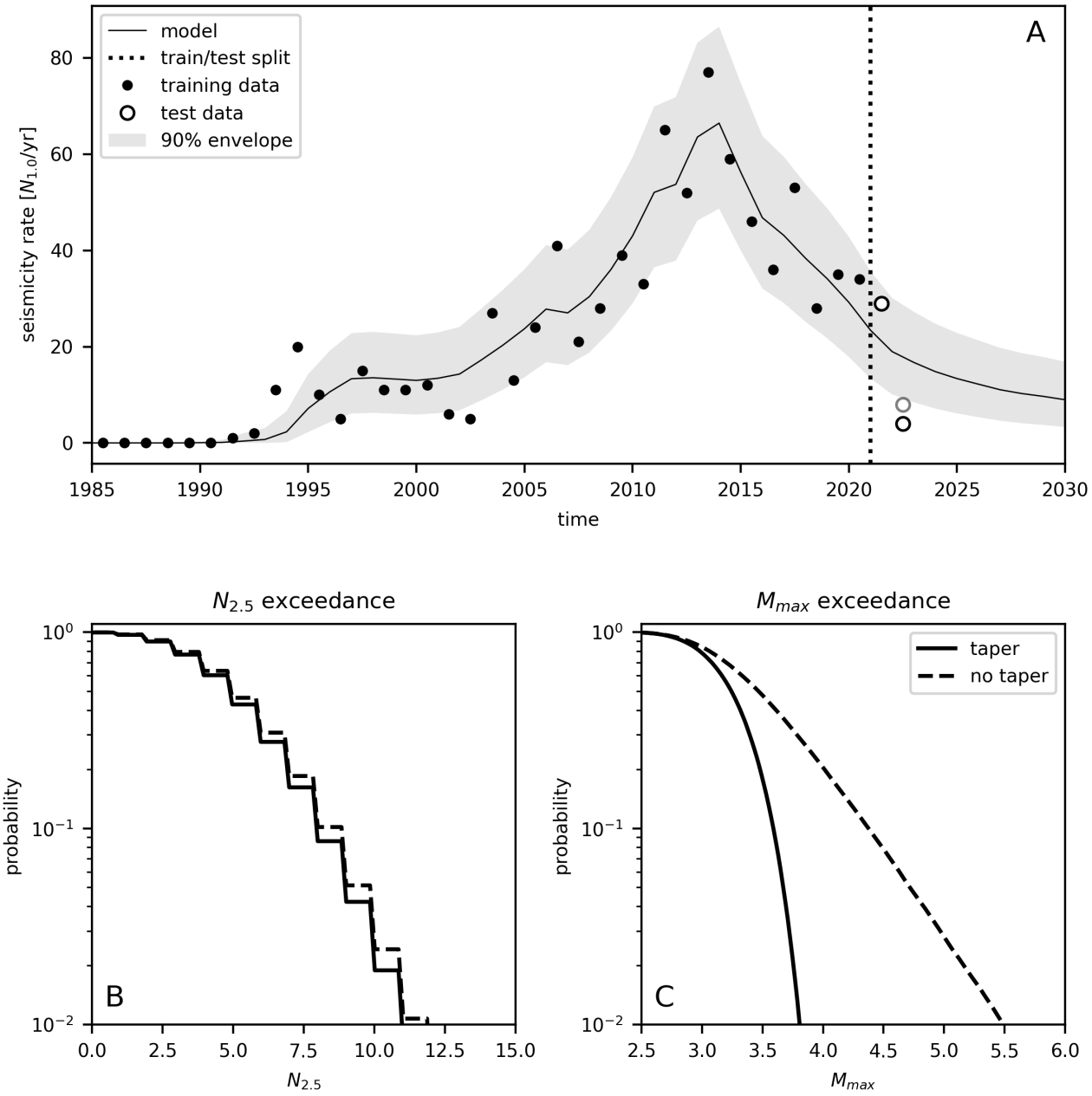


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Supporting Information for "Physics-based forecasting of induced seismicity at Groningen gas field, the Netherlands: post hoc evaluation and forecast update"

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Detailed description of *Dempsey and Suckale* (2017) seismological model

The Dempsey & Suckale (2017) forecast model, hereafter referred to as DS17, was used to estimate the possible maximum magnitude at Groningen, $M_{\max}$. This quantity was defined probabilistically as the largest earthquake resulting from hypothetical extraction scenarios at Groningen between 2017 and 2025. The model contained parameterised descriptions of (i) the initial tectonic stress field at Groningen, (ii) a poroelastic model that links gas production to fault loading, and (iii) simulations of earthquake source triggering and propagation.

Geomechanical model

The initial tectonic stress within the Groningen reservoir is poorly constrained. In the DS17 model, the tectonic regime was assumed to be extensional and was thus parameterized by a vertical maximum principle stress, σ_1, and two horizontal principle stresses, σ_2 and σ_3. The magnitude of σ_1 was calculated through depth integration of a constant overburden density (2500 kg/s) to the average reservoir depth (3 km).

The magnitude of the minimum horizontal stress, σ_3, is unknown. For an extensional system, it must be less than σ_1. However, it is presumed to not be smaller than a theoretical lower limit, σ_{crit}, imposed by frictional equilibrium for well-oriented shear cracks (Jaeger et al., 2009), given:

$$\sigma_{\text{crit}} = (\sigma_1 - p_0)\left[f + \sqrt{1+f^2}\right]^{-2} + p_0, \tag{1}$$

where p_0 is the initial pressure in the reservoir ($\sim$35 MPa in 1958.)

Given that σ_3 is in the range σ_{crit} to σ_1, its value can be parameterized by $h = (\sigma_3 - \sigma_{\text{crit}})/(\sigma_1 - \sigma_{\text{crit}})$. Thus h varies between 0 and 1 with $h = 0$ denoting a critical stress state, and $h = 1$ an isotropic stress state.

The magnitude of the maximum horizontal stress, σ_2, is assumed to lie between σ_1 and σ_3, and is parameterized by $H = (\sigma_2 - \sigma_3)/(\sigma_1 - \sigma_3)$.

Finally, the angle between σ_3 and north denotes the rotation of horizontal stresses (main text, Fig. 2a). Mathematically, it is applied as a rotation around a vertical axis

$$S = R(\theta)^T \, S_D \, R(\theta), \tag{2}$$

where S_D is a diagonal matrix containing the principle stresses, S is the stress tensor in x-y coordinates, and R is the rotation matrix.

Rotation of the stresses determines which faults are most likely to be well-oriented in the stress field for early reactivation (main text, Fig. 2a). Thus, the regional stress field was determined by four parameters: $[f, h, H, \theta]$. It was assumed to be homogeneous throughout the field.

Poroelastic fault loading

In the DS17 model, the depth-averaged spatio-temporal pressure distribution (main text, Fig. 2a) in the reservoir, $\Delta p(\mathbf{x}, t)$, was derived from a two-component (water and gas) reservoir simulation. The simulations accounted for historic gas extraction and were history matched to pressure and subsidence data. Three future depressurization scenarios were calculated for gas extraction scenarios averaging 21, 27 and 33 bcm/yr between February 2017 and 2025.

A poroelastic model was used to link depressurization to stress changes that triggered earthquakes. The model invokes the embedded ellipse conceptualization of Segall & Fitzgerald (1998), wherein depressurization induces a contraction of an elastically decoupled reservoir. The reservoir later recovers its original shape and true elastic coupling through the application of an extensional strain within the reservoir and subsidence above (main text, Fig. 2c).

This poroelastic model was applied locally, using the pressure change to calculate horizontal stress change by $\Delta\sigma_h(\mathbf{x},t) = A\Delta p(\mathbf{x},t)$ where h=2 or 3. The parameter, A, is a poroelastic stress path coefficient (Segall & Fitzgerald, 1998; Zoback, 2010). As a result, negative pressure changes (depressurization) cause the horizontal stresses to decrease, pushing the system closer to a critical extensional stress state. The term $\Delta\sigma_h(\mathbf{x},t)$ was applied as a modification to the two principle horizontal stresses in a homogeneous background tectonic stress

$$S_D(\mathbf{x},t) = \underbrace{\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}}_{\text{initial tectonic stress}} + A \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & \Delta p(\mathbf{x},t) & 0 \\ 0 & 0 & \Delta p(\mathbf{x},t) \end{bmatrix}}_{\text{poroelastic modification}}. \tag{3}$$

Thus, the regional stress field evolves in space and time as a function of depressurization.

Faults embedded in the stress field experience a time and space varying load (main text, Fig. 2b). In DS17, faults were treated as a straight line elements with the ith fault having normal vector, $\hat{\mathbf{n}}_i$, that depends on fault strike, φ_i, and dip, ϕ_i, as well as a length, L_i. The stress field imposes a traction force on the fault, $\mathbf{t}$, which in turn can be subdividing into normal, σ_n, and shear stress, τ, components.

$$\mathbf{t}(s,t) = S(\mathbf{x}(s),t)\hat{\mathbf{n}}_i, \qquad \sigma_n(s,t) = \hat{\mathbf{n}}_i \cdot \mathbf{t}(s,t), \qquad \tau(s,t) = |\mathbf{t}(s,t) - \sigma_n(s,t)\hat{\mathbf{n}}_i|, \tag{4}$$

where s is a coordinate along the strike of the fault.

DS17 earthquake physics simulator

The DS17 model was built using the induced earthquake physics simulator described in Dempsey & Suckale (2016). In that simulator, a fault was abstracted as a 1D line segment. The initial shear stress was heterogeneous along the fault ($\tau_0(s)$) with a mean value that depended on fault orientation in the regional stress field. The spatial correlation properties of $\tau_0(s)$ were assigned values consistent with seismological observations. Under tectonic loading, the simulator would output earthquakes with a Gutenberg-Richter MFD (Dempsey et al., 2016) and a Poissonian distribution of event times.

Model earthquakes were triggered by applied stress and strength perturbations. The spatio-temporal change of shear and normal stress on the fault ($\Delta\tau(s,t)$, $\Delta\sigma_n(s,t)$) were derived from the previously described poroelastic depressurization model. Mohr-Coulomb strength on the fault was the product of effective normal stress, $\sigma_n(s,t) - p(s,t)$, and the same friction coefficient, f, used to parameterize the stress state. An earthquake was triggered on the fault at the first location where stress exceeded strength across some minimal rupture patch size for nucleation. Once the earthquake was triggered, two rupture fronts propagated in either direction along the fault, away from the nucleation site. Continued propagation was dependent on strength drop and stress transfer, approximated by a simplified 1D crack energy model (main text, Fig. 2d). Once the rupture size was known, an earthquake magnitude was calculated using the total rupture length and average stress drop.

An ensemble of possible earthquakes was simulated for the 325 longest faults in Groningen (main text, Fig. 1a), with a minimum fault length of 2 km. Interactions between earthquakes on separate faults was not considered. The total simulated seismicity was used to define a time-varying seismicity rate for the field, $\lambda(t)$.

Empirical MFDs at Groningen

The unbounded GR distribution assumes that earthquake magnitudes, M, follow a power-law distribution with slope parameter, b. The exceedance probability that an event, M_i, will have a magnitude larger than M is given:

$$P(M_i \geq M) = 10^{-b(M-M_m)}, \tag{5}$$

where M_m is the minimum magnitude considered.

Bourne & Oates (2020) introduced the GR distribution modified with the exponential taper model of Kagan (2002). Its exceedance probability is given by:

$$P(M_i \geq M) = 10^{-b(M-M_m)} \exp\left[-\left(10^{1.5(M-M_C)} - 10^{1.5(M_m-M_C)}\right)\right]. \tag{6}$$

The parameter M_C is the corner magnitude where rollover of the MFD starts to occur. It is related to the exponential taper parameter, ζ, from Bourne et al. (2018) by

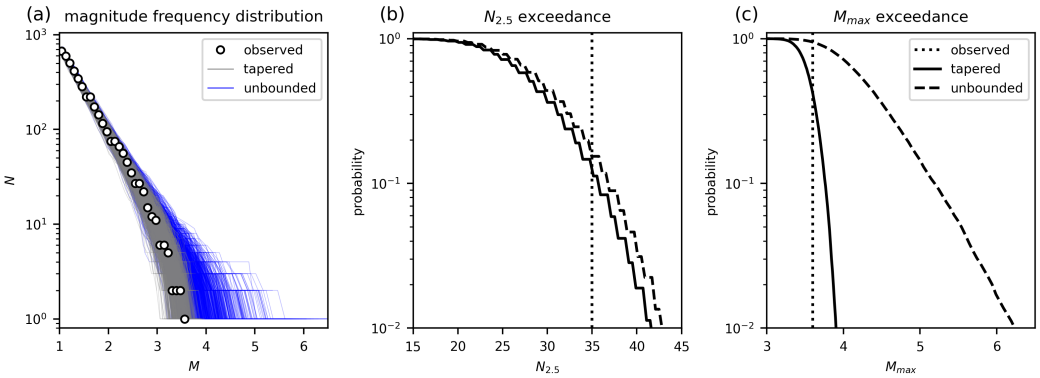


Figure 1. Summary of the Groningen magnitude frequency distribution (MFD) up to February 2017 and credible model fits. (a) Observed MFD (circles) and 1000 synthetic catalogs constructed from tapered (grey) and unbounded (blue) Gutenberg-Richter distributions. For the synthetic catalogs in (a), exceedance probability curves for (b) the number of events M 2.5 or larger, and (c) the largest event in the catalog. Both tapered (solid) and unbounded (dashed) models are shown with that actual outcome indicated as a vertical dotted line, i.e., 35 M≥2.5 events, the largest of which was a 3.6.

$M_C = M_m - \frac{1}{1.5}\log_{10}\zeta$. As $M_C \to \infty$, the exponential factor approaches one, recovering the unbounded GR distribution, Eq. (5). Bourne & Oates (2020) fit Eq. (6) to Groningen seismicity with M_m=1.5 and obtained 95% credible intervals for the parameters: 0.84< b <1.07 and 3.23< M_C <4.57.

Fig. 1(a) shows how both tapered and unbounded GR distributions can fit the MFD of Groningen seismicity prior to February 2017. For this comparison, we have selected b=0.9 for both models, M_C=3.5 for the tapered model, and generated 1000 synthetic catalogs to compare against the observations. The aleatoric variability in both models approximately bounds the data, although the tapered model has less spread in its distribution.

Exceedance probability curves indicate that both the tapered and unbounded GR models provide a credible fit for the number of M ≥2.5 events (Fig. 1b). The tapered model provides a better fit for the largest observed event with a probability to exceed M 3.6 of 0.43. In comparison, the unbounded model had 0.94 probability to exceed the observed maximum. Although the evidence suggests the tapered model is better supported by the data, in our view, it is not prudent to absolutely exclude the unbounded

model. This is mainly because of the large difference in $M_{\max}$ between the two models at the high exceedance limit, i.e., 1% likelihood to exceed M 3.9 for the tapered model and M 6.3 for the unbounded model.

Sensitivity of 2022 updated seismological model

Parameter sensitivity quantifies how certain model outputs may vary over plausible ranges of the model parameters. The key output whose sensitivity we are interested in is the prospective forecast and its exceedance curves. As examples for illustrating sensitivity, we have selected the stress state orientation, θ, and the poroelastic parameter, A. For each parameter, we constructed an ensemble of minimum misfit models, where each member of the ensemble had the sensitive parameter constrained to a narrow range. Then, we reconstructed the prospective forecast for each ensemble member. For example, θ was constrained in 10° intervals, and nine of these were constructed spanning the range [0,90°]. For A, we constructed nine evenly-spaced intervals in the range [0.8, 1.0]. Minimum misfit models, prospective forecasts and exceedance curves are shown in Figure 2.

The forecast model is not particularly sensitive to the parameter θ. Judged by eye, each model constructed over the tested range of θ provides a credible fit to the data, adequately capturing the pace of rise, peak and decrease in seismicity over the last 20 years (Fig. 2a). Furthermore, they all produce very similar prospective forecasts and exceedance curves for both tapered and unbounded GR models (Fig. 2B-C).

Certain elements of the forecast model are sensitive to the parameter A. First, the quality of model fit in the training period is more variable, although all models capture seismicity rise, peak and decline (Fig. 2d). There is more variability in both the rate forecast and exceedance curves for $N_{2.5}$. However, there is less variability in the $M_{\max}$ exceedance curves for both MFDs (Fig. 2E-F).

This exercise is useful to illustrate certain properties of history-matched forecasts. First, a model may achieve a reasonable match to the data when one of its parameters is knowingly assigned an incorrect value, e.g., θ. This happens because, during history matching, other parameters are able to take on values that compensate any misfit induced by the parameter change. This is parameter surrogacy (Doherty & Christensen, 2011) and it was a noted feature of the DS17 model.

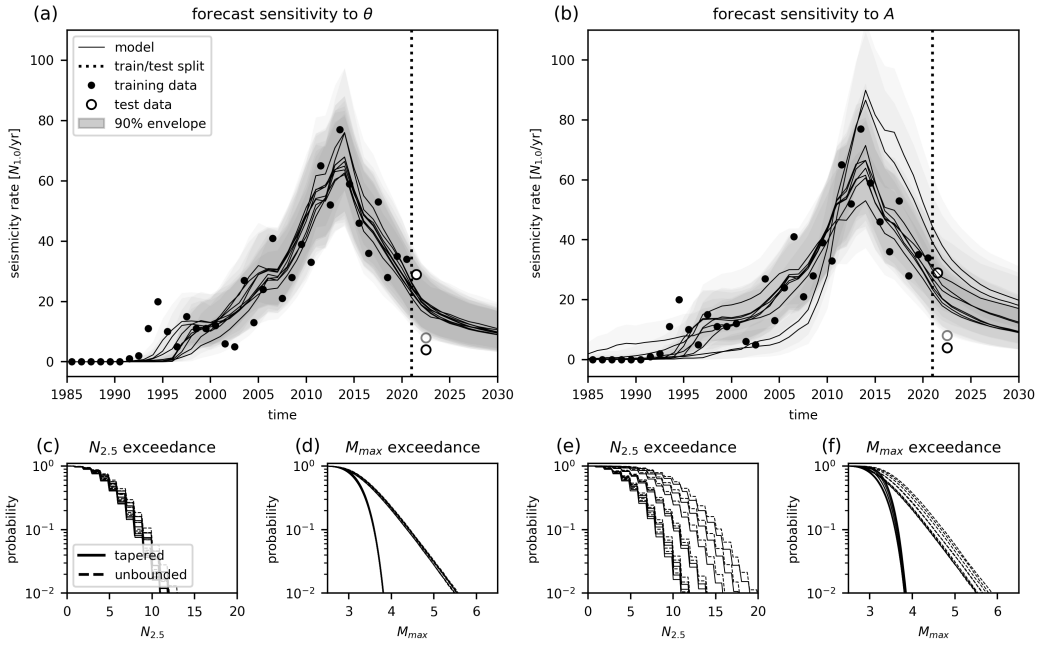


Figure 2. Summary of DS22 forecast sensitivity. (a) Nine best-fit models obtained when constraining θ to 10° intervals in the range [0,90]. Measured seismicity rate (circles) compared to best fit models (solid lines) with overlapping 90% aleatoric variability (grey envelopes). The greyed marker is the extrapolated event count for 2022 based on events in the first six months of the year. (b) Same as for (a) but instead nine best-fit models obtained when constraining A to evenly-spaced intervals in the range [0.8,1.0]. (c-d) Exceedance curves for prospective forecasts between 2021 and 2030 for the models in (a). (e-f) Same as for (c-d) but for the models in (b).

Second, a set of models with similar accuracy fitting the data have quite limited variability in their forecast outputs, regardless of the inferred correctness of those parameters. This, too, was a feature of the DS17 model and other forecasts that have sought to construct ensembles of well-fitting models (e.g., Richter et al., 2020; Smith et al., 2022).

Third, even seismicity rate models with dissimilar fits (Fig. 2d), some of which are relatively poor, may produce forecast outputs with little variability. In this case, convergence of the forecasts is induced by the upper limit in the tapered GR model or by the history match itself, which enforces seismicity rate decline on all models in the post-2014 peak period.

Description of depressurization scenario file

The file name is XY_PRF_Avg_OS2_CY.csv. The first two columns are x and y coordinates [m] of reservoir model output in the Groningen coordinate frame. Subsequent columns are depth-averaged pressure [bar] at the model output locations at the beginning of each year. The year is given by the column header. The spatiotemporal pressure evolution defined within this file was used to develop the forecast models described in the associated manuscript.

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